

✓ Lab 4 : Missing Data Handling in R

Dataset: Adult Census dataset (2000 row sample)

```
adult <- read.csv("adult.csv")
str(adult)
```

```
'data.frame':  32561 obs. of  15 variables:
 $ age      : int  90 82 66 54 41 34 38 74 68 41 ...
 $ workclass : chr  "?" "Private" "?" "Private" ...
 $ fnlwgt   : int  77053 132870 186061 140359 264663 216864 150601 88638 422013 70037 ...
 $ education : chr  "HS-grad" "HS-grad" "Some-college" "7th-8th" ...
 $ education.num : int  9 9 10 4 10 9 6 16 9 10 ...
 $ marital.status: chr  "Widowed" "Widowed" "Widowed" "Divorced" ...
 $ occupation : chr  "?" "Exec-managerial" "?" "Machine-op-inspct" ...
 $ relationship : chr  "Not-in-family" "Not-in-family" "Unmarried" "Unmarried" ...
 $ race      : chr  "White" "White" "Black" "White" ...
 $ sex       : chr  "Female" "Female" "Female" "Female" ...
 $ capital.gain : int  0 0 0 0 0 0 0 0 0 0 ...
 $ capital.loss : int  4356 4356 4356 3900 3900 3770 3770 3683 3683 3004 ...
 $ hours.per.week: int  40 18 40 40 40 45 40 20 40 60 ...
 $ native.country: chr  "United-States" "United-States" "United-States" "United-States" ...
 $ income     : chr  "<=50K" "<=50K" "<=50K" "<=50K" ...
```

```
install.packages("nanian")
install.packages("skmr")
library(nanian)
library(skmr)
```

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

also installing the dependencies ‘gridExtra’, ‘plyr’, ‘norm’, ‘visdat’, ‘viridis’, ‘UpSetR’

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

Attaching package: ‘skmr’

The following object is masked from ‘package:nanian’:

```
n_complete
```

```
set.seed(42)
```

```
n <- nrow(adult)
```

```
# Check column names and find the hours column automatically
hours_col <- names(adult)[tolower(gsub("[^a-z]", "", names(adult))) == "hoursperweek"]
```

```
if (length(hours_col) == 0) {
  stop("Could not find the hours-per-week column. Run names(adult) to check the column names.")
}
```

```
# Convert numeric columns
adult$age <- as.numeric(adult$age)
adult[[hours_col]] <- as.numeric(adult[[hours_col]])
```

```
# Create pool of row indices
all_indices <- 1:n
```

```
# Select unique indices
na_index <- sample(all_indices, 6)
all_indices <- setdiff(all_indices, na_index)

blank_index <- sample(all_indices, 6)
all_indices <- setdiff(all_indices, blank_index)

nan_index <- sample(all_indices, 4)
all_indices <- setdiff(all_indices, nan_index)

wrong_index <- sample(all_indices, 4)

# Introduce errors
adult$age[na_index] <- NA
adult$workclass[blank_index] <- ""
adult[[hours_col]][nan_index] <- NA
adult$age[wrong_index] <- 999

# Save dirty dataset
write.csv(adult, "adult_dirty_data.csv", row.names = FALSE)
```

```
sum(is.na(adult$age))
sum(is.nan(adult$hours_per_week))

x <- NULL
is.null(x)

sum(adult$workclass == "")
sum(adult$age == 999, na.rm = TRUE)

miss_var_summary(adult)
```

6
0
TRUE
6
4

A tibble: 15 × 3

variable	n_miss	pct_miss
<chr>	<int>	<num>
age	6	0.018 [4m4][24m]
hours.per.week	4	0.012 [4m3][24m]
workclass	0	0
fnlwgt	0	0
education	0	0
education.num	0	0
marital.status	0	0
occupation	0	0
relationship	0	0
race	0	0
sex	0	0
capital.gain	0	0
capital.loss	0	0
native.country	0	0
income	0	0

```
adult$age[adult$age == 999] <- NA
adult$workclass[adult$workclass == ""] <- "Unknown"

med_age <- median(adult$age, na.rm = TRUE)
adult$age[is.na(adult$age)] <- med_age

adult <- adult[!is.nan(adult$hours_per_week), ]

sum(complete.cases(adult))
```

0

```
impute_median <- function(x){
  med <- median(x, na.rm = TRUE)
  x[is.na(x)] <- med
  return(x)
}

test <- c(10, 20, NA, 40, NA, 60)
impute_median(test)

adult$hours_per_week <- impute_median(adult$hours_per_week)
```

10 · 20 · 30 · 40 · 30 · 60

```
dirty <- read.csv("adult_dirty_data.csv", stringsAsFactors = FALSE)

# Find hours-per-week column
hours_col <- names(dirty)[
  tolower(gsub("[^a-z]", "", names(dirty))) == "hoursperweek"
]

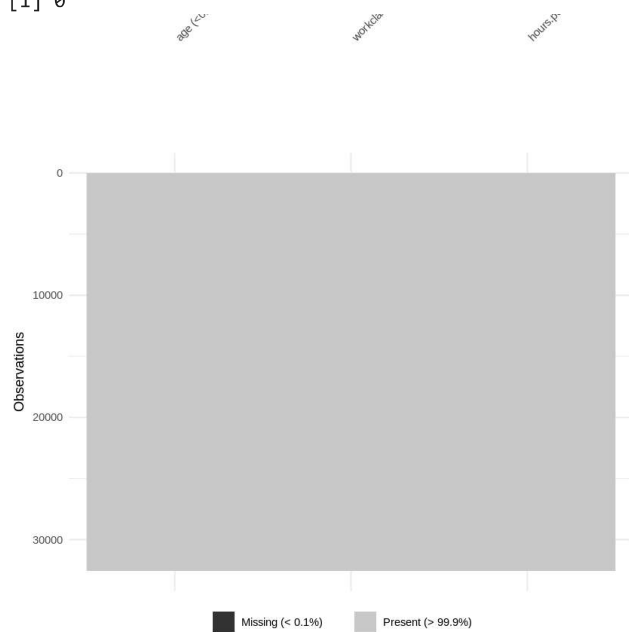
# Check error counts
before <- sum(is.na(dirty$age)) +
  sum(dirty$workclass == "") +
  sum(is.na(dirty[[hours_col]])) +
  sum(dirty$age == 999, na.rm = TRUE)

after <- sum(is.na(adult$age)) +
  sum(adult$workclass == "") +
  sum(is.na(adult[[hours_col]])) +
  sum(adult$age == 999, na.rm = TRUE)

print(before)
print(after)

# Visualize missing values separately
vis_miss(dirty[, c("age", "workclass", hours_col)])
```

```
[1] 20  
[1] 0
```



```
skim(adult)  
  
any(adult$age == 999)  
any(is.na(adult$age))  
any(adult$workclass == "")  
any(is.nan(adult$hours_per_week))  
  
write.csv(adult, "cleaned_adult_data.csv", row.names = FALSE)
```

	skim_type	skim_variable	n_missing	complete_rate	character.min	character.max	character.empty	character
	<chr>	<chr>	<int>	<dbl>	<lgl>	<lgl>	<int>	
1	character	workclass	0	NaN	NA	NA	0	
2	character	education	0	NaN	NA	NA	0	
3	character	marital.status	0	NaN	NA	NA	0	